Deep Phenotyping and Multimodal Data Integration for Early Prediction of Pregnancy Complications

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Context

Preterm Birth
<37 Weeks of Gestation
10.4% US Pregnancies

Gestational Hypertension
High Blood Pressure During Pregnancy
6-8% US Pregnancies

Fetal Growth Restriction
<10 Percentile Weight For Gestational Age
5-10% US Pregnancies

Figure 1. Common Pregnancy Complications. The biology underlying common obstetric disorders is unknown, and there are limited prevention and treatment options.

Cohort

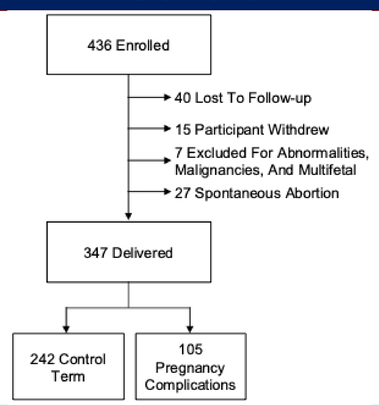

Figure 2. Cohort Composition. Participants enrolled at the first prenatal appointment at Magee-Womens Research Institute from February 2021 – March 2024. The cohort is reflects the population of Pittsburgh: 74.8% White, 13.1% Black; 83.9% Non-Hispanic.

Data Collection

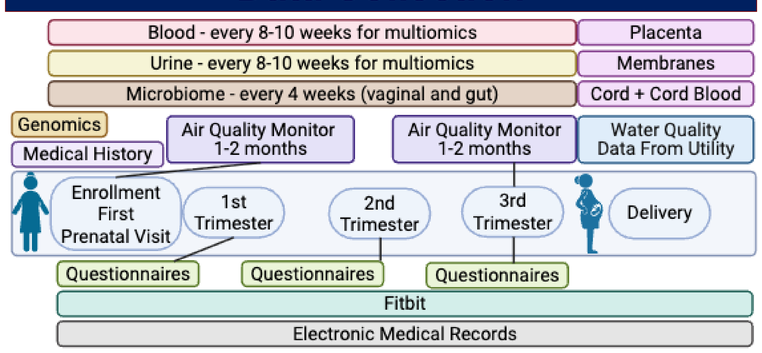

Figure 3. Timeline of Multidimensional Data Collection During Pregnancy. Microbiome samples are collected every 4 weeks, and blood and urine every 8 weeks, with additional biospecimens at delivery. Questionnaires are completed once per trimester and post-delivery. Environmental exposures are assessed via home air-quality monitoring (first two months after enrollment and early third trimester) and water-quality data from local utilities. Electronic health records and continuous Fitbit data are collected throughout pregnancy.

Obstetric Outcomes

Phenotypes	All n=347	Omics n=133
Term Control	242	56
Spontaneous Preterm Delivery (PTD)	15	13
Fetal Growth Restriction (FGR)	20	20
FGR with Hypertension Disorder during Pregnancy (FGR+HDP)	6	4
Gestational Hypertension (GHDP)	64	40

Table 1. Pregnancy Outcomes. Pregnancy outcomes by the entire cohort (n=347) and those for which multiomics was performed on biosamples (n=133). Multiomic processing was initially performed on 133 participants with the most complete temporally paired samples. Biosamples processed are: blood (lipidomics, metabolomics, proteomics, transcriptomics), urine (metabolomics, transcriptomics), placenta (lipidomics, metabolomics, proteomics, transcriptomics), cord blood (transcriptomics) and rectal and vaginal microbiome.

Complications	Yes	No	Total
Risk factor (Yes)	32	13	45
No Risk Factor	45	43	88
Total	77	56	133

Table 2. Presence of Risk Factors and Pregnancy Complications. The distribution of 133 participants with processed multiomics data by presence or absence of risk factors at enrollment and subsequent occurrence or non-occurrence of pregnancy complications. Risk factors included: history of pregnancy complications (e.g. preterm birth, gestational hypertension or preeclampsia), maternal age, pregravid BMI, or maternal comorbidities (diabetes or hypertension).

Goals

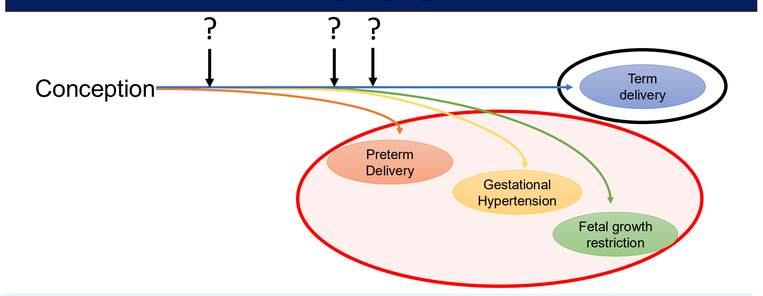

Figure 4. Graphical Representation of the Study's Goals. The major goals for this study are as follows:

1. Characterize the normal physiological process of pregnancy; identifying the major molecular transitions throughout pregnancy
2. Evaluate how people with pregnancy complications deviate from a healthy trajectory; defining mechanisms driving common obstetric syndromes
3. Identify the earliest possible point an individual patient deviates from a healthy trajectory; developing early, actionable diagnostic tools that enable timely, personalized interventions

FLT1/FSTL3 Community

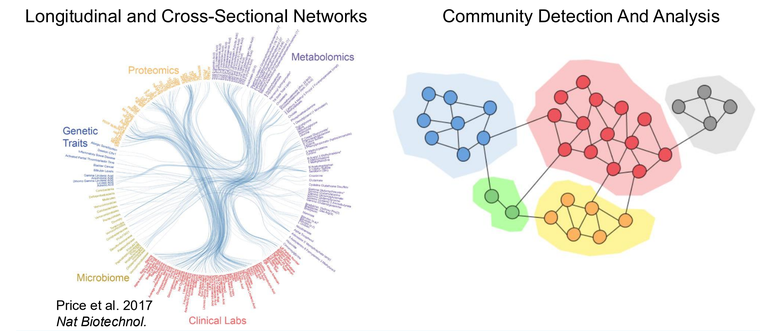

Figure 5. Define Network Changes Between Outcomes and Across Pregnancy. At each timepoint we will build intermodal networks using generalized linear models, adjusting for confounders. We will then perform community detection to define subnetworks. Changes in network structure will be evaluated across time and between term control and pregnancy complications at each timepoint.

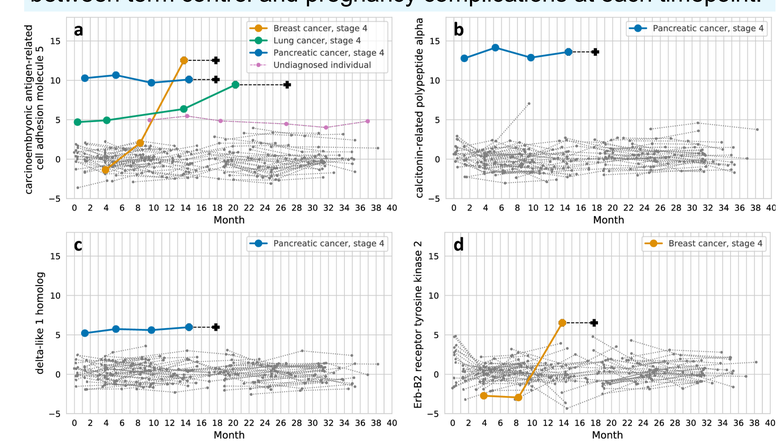

Figure 6. Temporal Modeling of Outliers. We will define the expected range for analytes using term controls [1, 99]. An analyte will be declared an outlier if it falls outside of this expected range. We will model changes in analyte level and outlier status over time to identify new biomarkers for pregnancy complications. Image displayed is a proof of concept from Magis et al. 2020, *Scientific Reports*.

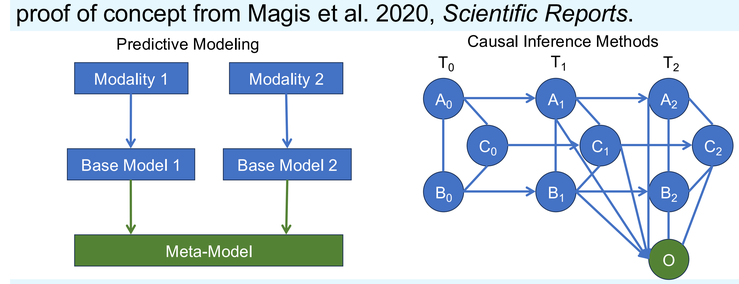

Figure 7. Developing Precision Medicine Models for Early Detection of Pregnancy Complications. The goal is to develop early, multimodal diagnostic platform to detect major pregnancy complications before symptoms arise. We will develop late-fusion ensemble machine learning models (left) and dynamic Bayesian networks (right). Model performance will be evaluated using accuracy, sensitivity, and lead time.